

Swathi Shree Narashiman

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EDUCATION

- **Indian Institute of Technology Madras** : B.Tech Electrical Engineering (Hons.) & Minor in Computing
CGPA: 9.74/10.0 Nov '22 – Present
- **DAV Group of Schools, Vellore** - Science Stream (CBSE)
Class 12: 498/500 **Class 10:** 493/500 Jun '20 – Jul '22

SCHOLASTIC ACHIEVEMENTS

- Recipient of S. Subramaniam Award (1st year), M. Saradha & Sankariah Scholarship (2nd year) for securing **highest CGPA** in B.Tech IIT Madras.
- One among the **100 Opjems Scholars** in India for the year 2024 (issued by O.P. Jindal Group)
- Secured All India Rank **759** in JEE Advanced 2022 and All India Rank **606** in JEE Main 2022
- Was issued **KVPY Fellowship** by IISc Bangalore for securing All India Rank **475** in KVPY-SX 2022

PUBLICATIONS

1. S. Hussein, J. Guan, **Swathi Shree Narashiman**, *et al.*, “3d shape reconstruction from autonomous driving radars,” in *The Thirty Sixth British Machine Vision Conference*, 2025
2. D. Panda, **Swathi Shree Narashiman**, J. K. Joe, *et al.*, *Fine-grained control over music generation with activation steering*, 2025. arXiv: [2506.10225 \[cs.SD\]](#)
3. **Swathi Shree Narashiman** and N. Chandrachoodan, *Alphazip: Neural network-enhanced lossless text compression*, 2024. arXiv: [2409.15046 \[cs.IT\]](#)
4. **Swathi Shree Narashiman**, V. A, D. Joshi, *et al.*, *Chiptlets on wheels: Review paper on holistic chiptlet solutions for autonomous vehicles*, 2024. arXiv: [2406.00182 \[cs.AR\]](#)

RESEARCH & EXPERIENCE

- **Summer@EPFL Research Internship** May '25 – Jul '25
Mentor: [Prof. Nicholas Flammarion](#), TML Lab, EPFL Lausanne, CH
 - Researched **Mechanistic Interpretability** in transformers, focusing on In-Context Learning and Induction Heads
 - Investigated theoretical guarantees and empirical results for **Markovian prediction** in Large Language Models
 - Established that transformers approximate finite-length Markov chains using weighted unigram and bigram counts
- **Adversarial Attacks in Large Language Models** Sep '24 – Jun '25
Mentor: [Prof. Nambi Seshadri](#), UC San Diego Remote
 - Investigated **error propagation** in neural networks via logit-space attacks on open-source LLMs
 - Analyzed the impact of injected noise and channel erasures on BLEU, Perplexity, and Semantic Similarity metrics
 - Verified that LLMs are robust to noise, while BLEU scores vary by $\sim 30\%$, preserving linguistic semantics [[Report](#)]
- **Music Interpretability via Activation Steering** Apr '24 – May '25
Club: [AI Club](#), Center for Innovation, IIT Madras Chennai, India
 - Used **steering** and **residual stream analysis** for controllable **fusion music generation** via generative models
 - Enabled genre, instrument, and pitch transfer efficiently by training $<0.005\%$ parameters on $<1K$ samples
 - Published results & findings on Fine-grained control over music generation on Arxiv [[Preprint](#)]
- **Summer@EPFL Research Internship** May '24 – Aug '24
Mentor: [Prof. Haitham Al Hassanieh](#), SENS Lab, EPFL Lausanne, CH
 - Selected among the **top 1.6%** globally to conduct research on **3D reconstruction** from **autonomous driving radars**
 - Enhanced the reconstruction using PCN and KL-based knowledge transfer, achieving a Chamfer distance below 7
 - Conference paper submitted & **accepted to the British Machine Vision Conference (BMVC) 2025**, Sheffield, UK
- **Neural Network-Enhanced Lossless Text Compression** Aug '23 – Sep '24
Mentor: [Prof. Nitin Chandrachoodan](#), IIT Madras Chennai, India
 - Leveraged LLMs in rank-based predictive compression, achieving $4.5\times$ compression on the **Gutenberg corpus**.
 - Analyzed diverse neural architectures coupled with information-theoretic frameworks like Huffman, Gzip etc.
 - Conducted empirical evaluations on domain-specific LLMs and published findings on arXiv [[Preprint](#)]

TECHNICAL PROJECTS

- **CS6170: Randomized Algorithms Course Project** Jan '25 – May '25
Instructor: [Prof. Yadu Vasudev, IIT Madras](#) | [Report](#) | [Video](#) Chennai, India
 - Explored **Online algorithms** for **AdWords** & **bipartite matching** under adversarial & stochastic models
 - Surveyed primal-dual & perturbation-based techniques including recent improvements over the classical $1 - \frac{1}{e}$ bound
- **EE6180: Advanced Topics in AI Course Project** (with [Immerso.ai](#) startup) Jan '25 – May '25
Instructor: [Prof. Pravin Nair, IIT Madras](#) | [Report](#) | [Github](#) Chennai, India
 - Trained Stable Diffusion models via **Textual Inversion** (< 1M params) to learn movie characters from 5-10 images
 - Designed prompt templates and token strategies to improve realism and control in few-shot generation
- **EE6150: Stochastic Modeling Course Project** Jan '25 – May '25
Instructor: [Prof. Avhishek Chatterjee, IIT Madras](#) | [Report](#) Chennai, India
 - Studied Glauber dynamics in sparse Ising models with near-linear mixing via graph decomposition
 - Examined Log-Sobolev methods and structural partitioning for efficient sampling in community detection

SKILLS

Python, C, C++, PyTorch, TensorFlow, LLMs, OpenCV, Git, Verilog, Xilinx Vivado, ARM, Arduino IDE, Altium Designer (PCB)

RELEVANT COURSEWORK

- **Machine learning & AI** : Advanced Topics in AI; Stochastic Modeling & Theory of Queues
- **Algorithms & Computing** : Randomized Algorithms; Quantum Algorithms; Computer Organization; Cryptography
- **Mathematics & Theoretical Foundations** : Linear Algebra, Convex Optimization, Probability, Control Systems, Information Theory, Multivariable Calculus, Series & Matrices

TEACHING EXPERIENCE

- **Teaching Assistant for Computer Organization** Jul '25 – Nov '25
Instructor: [Prof. Janakiraman Viraraghavan, IIT Madras](#) Chennai, India
 - Undergraduate course on Computer Organization: Evaluated examination scripts & conducted tutorial sessions.
- **Avanti Mentor** Nov '22 – Jul '23
Club: [Avanti Fellows, IIT Madras](#) Chennai, India
 - Provided one-on-one mentorship for underprivileged students to assist them in IIT-JEE preparation

POSITIONS OF RESPONSIBILITY

- **Project Lead of AI Club** Apr '24 – May '25
[AI Club, Center for Innovation, IIT Madras](#) Chennai, India
 - As a core team member, led a team of 40 enthusiasts in the development fusion music generation using AI
 - Presented at the **Open House** event, attracting 500+ attendees and multiple media outlets
- **Core member of EE Research Club** May '24 – present
[EE Research Club, IIT Madras](#) Chennai, India
 - Leading 30-member EE Research Club focused on cultivating a research mindset and building a scientific community
 - Ideated, organized, and hosted events to foster research engagement like socials, professor interviews, tech talks, etc.
- **Powertrain Engineer of Raftar Formula Racing** Apr '23 – Mar '24
[Raftar Formula Racing, Center for Innovation, IIT Madras](#) Chennai, India
 - Designed and tested key EV circuitry in the Low Voltage Subsystem of IITM's Formula Student electric race car
 - Team secured **Runners-up** position at **Formula Bharat 2024** with an international-level vehicle build
- **Press Correspondent** May '25 – present
[The Fifth Estate, IIT Madras](#) Chennai, India
 - Contributed articles to the official student press, covering diverse aspects of campus life and student affairs
- **Contingent** Oct '23 – Dec '23
[Inter IIT Tech Meet 12.0](#) Chennai, India
 - Proposed a chiplet-based solution to Jaguar Land Rover's High Prep problem as institute representative
 - Published on arXiv a cost-efficient SoC-to-chiplet design optimizing die size, interconnects, and architecture [[Preprint](#)]